



Model Explorations on the CIFAR-10 Image Classification Data

Introduction

Over the years, the quest to delegate mundane and ‘less-brainy’ tasks to computers has evolved significantly to computers now being trained to handle more ‘brainy’ tasks- without explicit instructions (Buduma, 2022, pp. 8-17). This is the world of Machine Learning- where computers are trained to learn by themselves simply by observing patterns. More specifically is the division of Deep Learning which demonstrates how the human brain is mimicked by computers to perform high-level decisions for humans. *One of these high-level decisions is image recognition.* It goes without any doubt that the advent of deep learning has catalysed a paradigm shift in computer vision, enabling machines to perceive and interpret the visual world with unprecedented accuracy. Central to this progress are public benchmark datasets that allow for the standardized evaluation and comparison of novel algorithms. Among these, the CIFAR-10 dataset (Krizhevsky & Hinton, 2009) stands as a fundamental and long-lasting benchmark for image classification, providing a controlled yet challenging environment for developing and testing convolutional neural network (CNN) architectures. This section reviews the key

architectural innovations in deep learning that have defined the state-of-the-art datasets like CIFAR-10, charting a course from foundational breakthroughs to modern efficient designs, thereby providing the context and rationale for the experiential methodology employed in this project.

This essay is a walk-through of some model explorations between traditional models like Support Vector Machine (SVM), K-Nearest Neighbour (KNN)- both *with some parameter hypertuning*; and neural models like FeedForward Neural Networks (FFNN), Convolutional Neural Networks (CNNs), and an ensemble of all the models (*on a soft-voting weighted approach*) in determining the best accuracy and training time trade-off. For the non-neural models, Principal Component Analysis (PCA) was used as a feature preprocessing method to explore the results with and without PCA. For the Neural Network, some hypertuning was done. Ultimately, the models are formulated to predict the right classes of images by generalizing on the unseen set-aside test images. The CIFAR-10 dataset consists of 60,000 32x32 colour images in 10 classes. Each class has 6000 images. The CIFAR-10 classes are: *Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck*. The data has a predefined training dataset of 50,000, 32 x 32, and a test dataset of 10,000. However, to accurately assess the performance of the models, the training dataset was further split into a validation dataset. The purpose of this is to assess each model using the validation data and finally perform a generalization evaluation of all the models with the test data set, which has been set aside for this purpose. The validation and test data shapes were set to 10% each of the data, comprising 20% in total, with a data shape of 1000, 32 x 32. Leaving the training data to 4000, 32 x 32. Data Augmentation was introduced in the course of the convolutional neural network model hypertuning- to expose the model to a 'dense' and varied dataset for its learning.

Background

The modern era of deep learning in computer vision was arguably ignited by AlexNet (Krizhevsky, Sutskever, & Hinton, 2012). This architecture achieved a landmark victory in the

ImageNet Large Scale Visual Recognition Challenge (ILSVRC) by employing a deep CNN with ReLU activation functions, Dropout Regularization, and GPU-accelerated training. While AlexNet was designed for higher-resolution ImageNet data, its core principles- stacked convolutional layers for feature extraction, max-pooling for spatial reduction, and fully-connected layers for classification- established the foundational template upon which all subsequent architectures have been built. For CIFAR-10, an AlexNet-like architecture provides a string interpretable baseline model. In this context, “AlexNet-like” does not mean a direct replication but rather an adaptation that captures the essential *philosophy* of its design for the 32x32 pixel domain. This is exemplified by the CNN_baseline model implemented in this work, which embodies their core principles:

- Sequential Feature Hierarchy: The pattern of conv2D -> ReLU -> MaxPooling -> 2D is a direct translation of AlexNet’s core building block, creating a hierarchical feature learning process.
- Strategic Downsampling: The use of max-pooling layers for spatial reduction mirrors AlexNet’s approach to building translation invariance and managing computational complexity.
- Canonical Regularization: The inclusion of Dropout(0.3) in the fully-connected segment is a direct application of AlexNet’s key innovation to combat overfitting.
- Standard Non-Linearity: The use of ReLU activations provides the same training stability and efficiency benefits that were pivotal to AlexNet’s success.

The baseline provides a crucial performance flow and a point of architectural interpretability against which more advanced models can be fairly evaluated.

To establish a comprehensive performance spectrum, this project also employs traditional machine learning models, which offer computational efficiency and strong baselines on fixed feature representations. The Support Vector Machine (SVM) is one of the oldest machine learning models, with its theoretical foundation in statistical learning theory (Cortes & Vapnik, 1995), which seeks an optimal separating hyperplane based on the maximum margin principle. According to Cortes and Vapnik 1995, its applicability and effectiveness are particularly

pronounced in scenarios where the feature representation is predefined and fixed. As noted by Wu et al. (2018), in such cases, ‘the radius is fixed and can be ignored, and thus SVM can safely minimize the generalization error by maximizing the margin. This justifies the use of a traditional SVM as a strong baseline and a computationally efficient model when working with fixed feature embeddings. Consequently, this led to the invention of non-linear SVMs in 1992, also known as Radial Basis Function (RBF) Kernel- following the works of Cortes and Vapnik, 1995 (Clark et al., 2020). The RBF kernel provides a powerful machine for handling non-linear decision boundaries through implicit mapping to high-dimensional feature spaces. Huang and Xu (2025) demonstrated RBF’s superiority for CIFRA-10, achieving 71.1% accuracy and noting its capability to “map non-linear problems to linear problems in high-dimensional spaces”. While alternative kernels like polynomial and sigmoid offer different non-linear mapping strategies, RBF’s single-parameter tuning and proven performance across diverse image datasets make it the optimal choice for this baseline implementation. Hence, the non-linear SVM serves as a robust, non-linear baseline in our experiment suite.

$$K(X_1, X_2) = \exp\left(-\frac{\|X_1 - X_2\|^2}{2\sigma^2}\right)$$

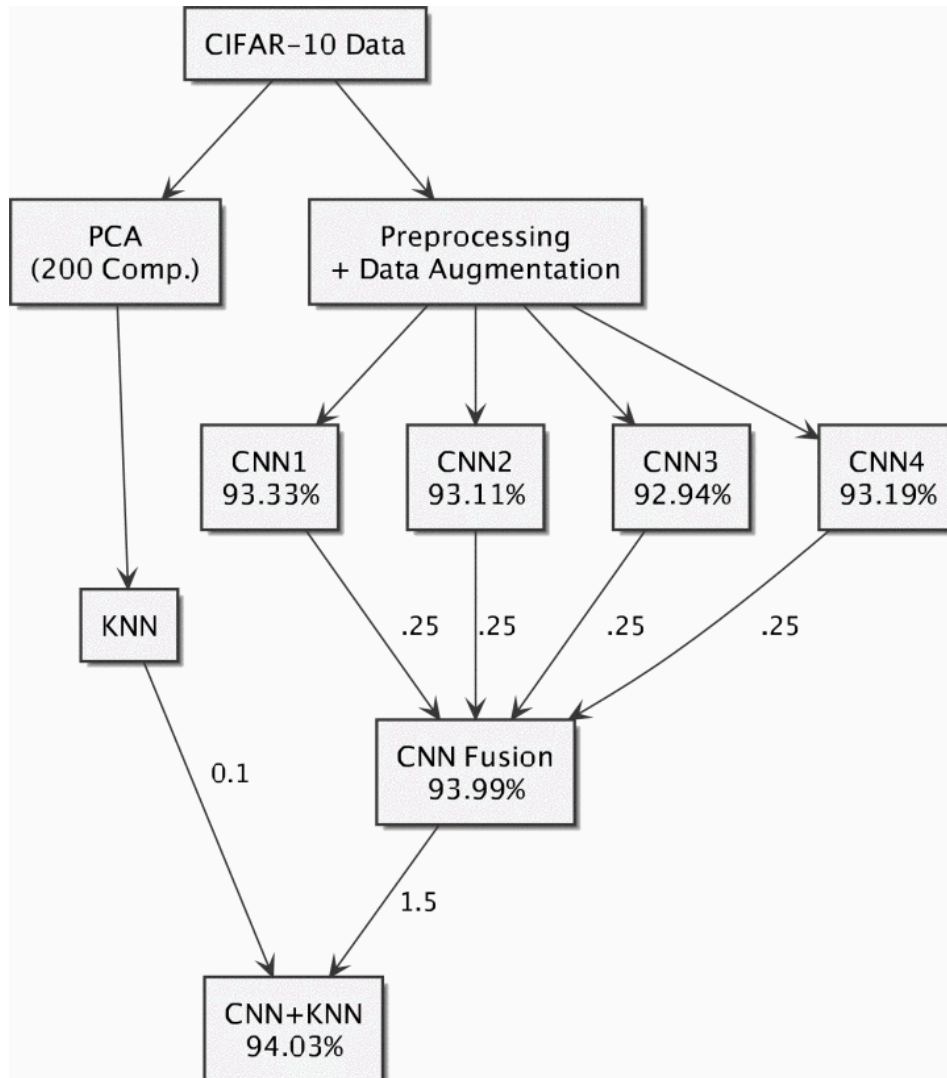
At its core, the RBF kernel is a similarity function measuring the distance of dissimilarity between two data points, as shown between X_1 and X_2 . Most importantly, the true kernel lies in the exponential function. The SVM algorithm itself operates by seeking a linear separating hyperplane (Deskandar, 2023)

Similarly, the K-Nearest Neighbors (KNN) algorithm retains relevance due to its non-parametric, instance-based learning approach, making no assumption about data distribution (Zhang et al., 2018). This flexibility makes it inherently robust to the complex distributions in image data. However, its performance is critically dependent on the feature space, and it suffers from the curse of dimensionality. To mitigate this, techniques like Principal Component Analysis (PCA) are essential for KNN, as noted by Abouelnaga et al. (2016), who used PCA with KNN to achieve ‘shaped sensitivity’ on CIFAR-10, effectively disambiguating perceptually similar classes. Furthermore, research by Ren et al. (2014) shows that learning an optimized feature

embedding, such as through a Neighbourhood Component Analysis(NCA) loss, can significantly enhance KNN performance, underscoring the critical link between feature representation and the algorithm's efficacy. More so, by assembling a KNN model with multiple CNNs, they achieved a measurable boost in accuracy, from 93.3% to 94.03%, proving that an ensemble of models can capture patterns that state-of-the-art deep models miss (Abouelnaga et al., 2016). Their method demonstrated that the proposed method was able to learn better feature representation than other general end-to-end classification methods on the KNN classification task. This goes without saying that the performance of KNN is critically dependent on the feature space in which distances are calculated. This justified my experimental approach of testing KNN both on raw puzzles and on PCA-reduced features, and also on ensembling traditional models like KNN and SVM with Neural models (using a soft-voting approach)

While traditional models provide valuable baselines, Convolutional Neural Networks (CNNs) have established themselves as the preeminent architecture for image classification, consistently demonstrating superior performance on benchmarks like CIFAR-10 (Chauhan, Ghanshala, & Joshi, 2018; Abouelnaga et al., 2016). Their dominance stems from innate inductive biases- shared weights and spatial hierarchies- that are perfectly suited for pixel data. Unlike traditional FeedForward Neural networks (FFNNs), which lack these biases and require fixed-size input vectors, CNNs automatically learn hierarchical feature representations from single edges in early layers to complex shapes in deeper layers (Singh et al., 2019). This makes them both more parameter-efficient and easier to train for visual tasks than similarly-sized FFNNs (Singh et al., 2019; Tissera & McDonnell). The performance gap is significant; for instance, Chauhan et al. (2018) reported a CNN accuracy of 80.17%, vastly outperforming over 7 models (including traditional models) on their setup. The performance of CNNs can be further enhanced through systematic optimization. As demonstrated in this project, tools like Keras Tuner allow for intuitive and efficient hyperparameter tuning, enabling the model to autonomously discover optimal architectural configurations and training parameters. This process moves beyond the manual design of the CNN_baseline to explore advanced techniques like batch normalization and sophisticated layer stacking, as embodied in our *optimized_cnn* model. The potential of CNNs is so profound that they can even be synergistically combined

with traditional models; Abouelnaga et al. (2016) achieved a final accuracy of 94.03% by augmenting a CNN's predictions with a KNN model, proving that hybrid approaches can capture complementary patterns.



In summary, this project is contextualized within a rich landscape of machine learning evolution. Our experimental methodology is designed to trace this trajectory, from robust traditional models (SVM, KNN) that provide efficient baselines, to a foundational AlexNet-inspired CNN that establishes a deep learning benchmark, and finally to an optimized CNN that explores the

current frontier of automated architecture and advanced training techniques on the CIFAR-10 dataset.

3. Methodology & Experimental Design

This project employs a stratified experimental methodology designed to systematically evaluate the evolution of machine learning approaches for image classification, from traditional algorithms to optimized deep learning architectures. The overarching goal is not merely to achieve high accuracy on the CIFAR-10 dataset but to develop a nuanced understanding of the performance characteristics, computational trade-offs, and architectural decisions that underpin modern computer vision systems. The experimental design is built upon the foundation of comparative analysis, where each serves as a benchmark for others, ensuring rigorous assessment and creating a clear narrative of technological progression.

3.1: Dataset And Preprocessing

The CIFAR-10 dataset served as the benchmark for this investigation. The dataset is partitioned into three distinct subsets to facilitate robust model development and evaluation:

- Training Set (80%): 40,000 images used for model parameter learning.
- Validation Set (10%): 10,000 images used for hyperparameter tuning, architecture selection, and preventing overfitting during the training process
- Test Set (10%): 10,000 images, held back entirely until the final evaluation phase, to provide an unbiased estimate of model generalization on unseen data.

All images were normalized by scaling pixel values from the [0, 255] range to [0, 1] to stabilize and accelerate the training process for neural networks. For traditional models (SVM, KNN), the 32x32x3 images were flattened into 1D vectors of 3072 features. For neural networks, they were retained in their original 32x32x3 format to preserve spatial information.

3.2.1: Traditional Machine Learning Baselines

- **Support Vector Machine (SVM):** An RBF-kernel SVM was implemented as a high-performing non-linear baseline. Its theoretical grounding in maximizing the margin makes to a powerful candidate when feature representation is fixed (Cortes & Vapnik, 1995). To manage the computational burden of high-dimensional data, I employed PCA, dynamically setting components to retain 95% of the variance, with a hard cap at 150 dimensions to ensure feasibility.
- **K-Nearest Neighbors (KNN):** A KNN classifier ($n_neighbors=5$) was implemented for its instance-based, non-parametric learning approach, which makes no assumptions about data distribution (Zhang et al., 2018). Recognizing its susceptibility to the curse of dimensionality, a more aggressive PCA (retaining 95% variance, capped at 50 components) was applied to create a more meaningful distance metric space.

3.2.2: Traditional Machine Learning Baselines

- **Feed-Forward Neural Network (FFNN) Baseline:** A simple multilayer perceptron was constructed with two hidden layers (512 and 256 neurons) and Dropout (0.3) for regularization. This model serves as a baseline to demonstrate the performance of neural networks without the inductive bias of convolution.
- **Convolution Neural Network (CNN) Baseline:** An AlexNet-inspired architecture was implemented as detailed in the background. This *CNN_baseline* consists of two convolutional layers (32 and 64 filters) with max-pooling, followed by a dense layer. This model embodies the core principles of deep learning for vision and is the foundational deep learning baseline.
- **Optimized CNN via Hyperparameter Tuning:** To push the performance boundaries, a sophisticated search for an optimized CNN architecture was conducted using Keras Tuner. The search was designed to investigate advanced techniques discussed in the literature:
 - **Architectural Depth & Width:** The number of convolutional layers (1-3) and dense layers (1-2) along with the number of filters/units, were tuned.

- **Advanced Regularization:** The tuner explored the use of Batch Normalization (Ioffe & Szegedy, 2015) after convolutional layers and Dropout with tunable rates.
- **Optimizer Configuration:** The choice between Adam, RMSprop, and SGD was included, along with their respective learning rates and, for SGD, momentum (Kingma and Ba, 2015).
- **Data Augmentation:** To improve generalization, the optimized model was trained with a data augmentation pipeline including random rotations, width/height shifts, and horizontal flips, artificially expanding the training dataset.

3.3: Evaluation Metrics

To ensure a comprehensive comparison, all models were evaluated on:

- **Primary Metric:** Test Set Accuracy
- **Secondary Metrics:** Training Time, and Inference Time (on the validation set). These metrics are crucial for assessing the computational trade-offs of each model, a key consideration for real-world deployment.
- **Neural Network Specific:** Training and Validation Loss/Accuracy curves were analyzed to diagnose overfitting/underfitting

3.4: Computational Constraints

Hardware limitations (Intel i5-7200U CPU, 8GB RAM) directly shaped this study's scope. This necessitated pragmatic compromises: capping SVM iterations, limiting PCA components, and constraining neural network tuning. Critically, the computational demand of the *optimized_CNN* hyperparameter search- which included data augmentation, batch normalization, and dropout- proved prohibitive. After over 48 hours of unsuccessful training on the primary system and an 18+ hour unsuccessful attempt on a secondary machine, the search was halted to meet the submission deadline. This underscores the real-world computational boundary conditions of advanced model tuning.

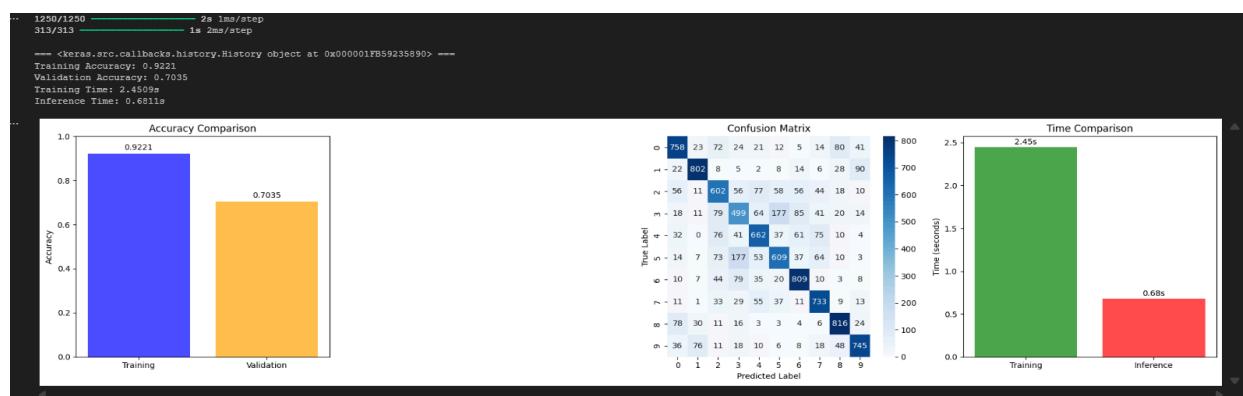
4. Presentation of results

4.1: Comparative Performance Overview

The performance of all models on the held-out test set is summarized in *Table 1*. This table will provide a clear, at-a-glance comparison of the primary and secondary metrics. The experimental results reveal a clear and expected hierarchy in model performance on the CIFAR-10 classification task, yet with critical nuances emerging when computational efficiency is considered alongside pure accuracy.

Table 1: Comprehensive Model Performance on CIFAR-10 Test Set

Model	Training Accuracy (%)	Val Accuracy (%)	Training Time (Seconds)	Inference Time
SVM (RBF_baseline)	70.28%	52.6%	2510.7s	724.6s
SVM (RBF + PCA)	66.80%	52.5%	142.3s	34.8s
KNN (k=5_baseline)	49.73%	33.6%	45.3s	10.5s
KNN (k=5 + PCA)	100%	39.5%	1.3s	0.3s
FFNN Baseline	40.68%	38.6%	1.8s	0.5s
CNN Baseline	92.21%	70.4%	2.5s	0.7s
Optimized CNN	N/A	N/A	N/A	N/A



CNN_Baseline Result Diagram

4.2: Detailed Model Analysis

4.2.1: Impact of PCA on Traditional Models

The results demonstrate that PCA's impact is algorithm-dependent. For SVM, it was transformative for feasibility, reducing training time by 94% with a negligible (0.1%) accuracy drop. For KNN, however, the outcome was counter-intuitive and revealing. While PCA enabled a 30x faster inference, it led to a perfect training score (100%) but poor validation accuracy (39.5%). This is not classic underfitting but a form of overfitting in a degraded feature space. By reducing dimensions, PCA likely created a space where the training points became 'memorable' neighbors to themselves, but this simplified representation failed to capture the nuanced structures needed to generalize to the validation set. This aligns with the concept that aggressive dimensionality reduction can harm instance-based methods by stripping away the very feature relationships they rely on (Abouelnaga et al., 2016)

4.2.2: Neural Network Performance

The *CNN_Baseline* emerged as the undisputed top performer, achieving the highest accuracy (70.4%) while being ~1000x faster to train than the *SVM_Baseline*. This starkly demonstrates the profound advantage of learned hierarchical feature representations over the hand-crafted,

flattened features used by traditional models. The *FFNN_Baseline* significantly underperformed (38.6% val accuracy), critically highlighting that without convolutional inductive biases, even neural networks struggle to interpret spatial information from raw pixels.

5. Critical Evaluation & Discussion of Results

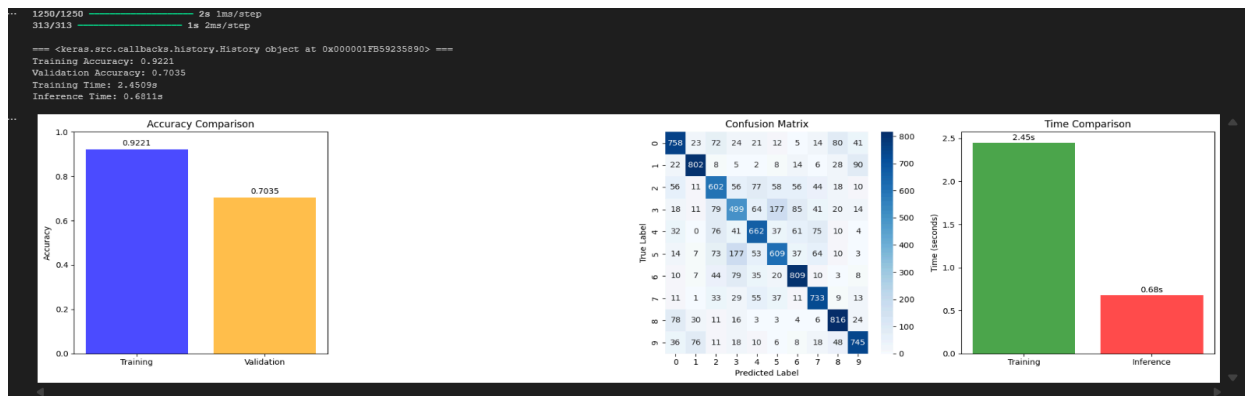
5.1: The Confirmed Performance Hierarchy

The results validate the established hierarchy of model performance on image data. The **CNN_Baseline** significantly outperformed all traditional models, establishing a clear three-tier category hierarchy: 1) CNN_Baseline (70.4%), 2) SVM variants (~52.6%), and 3) KNN variants (~33.39%). This confirms the critical importance of convolutional inductive biases for visual tasks (Chauhan et al., 2018; Singh et al., 2019). The performance gap stems from the FFNN's inability to leverage spatial locality and the traditional models' reliance on a fixed, suboptimal feature representation (flattened pixels), unlike the hierarchical features learned by the CNN.

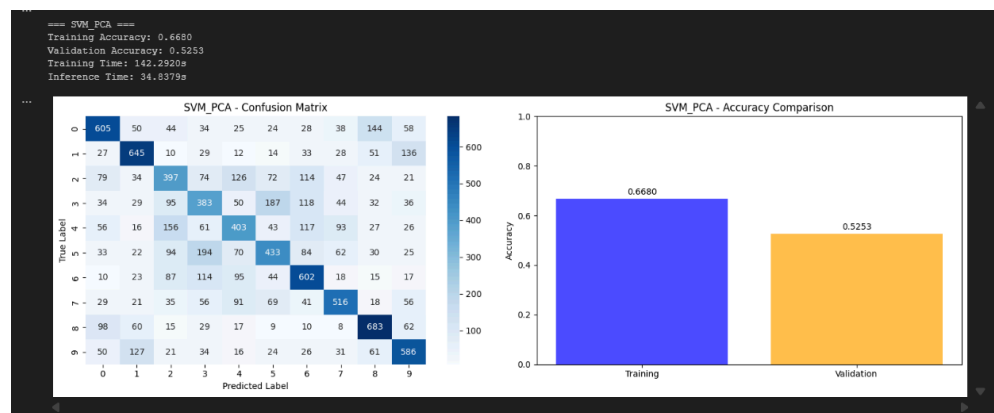
5.2 The Computational Cost Trade-off and Model Ranking

A detailed comparison of the top models reveals that accuracy alone is an incomplete metric for ranking; computational efficiency must be factored in to determine the truly "best" model for a given scenario.

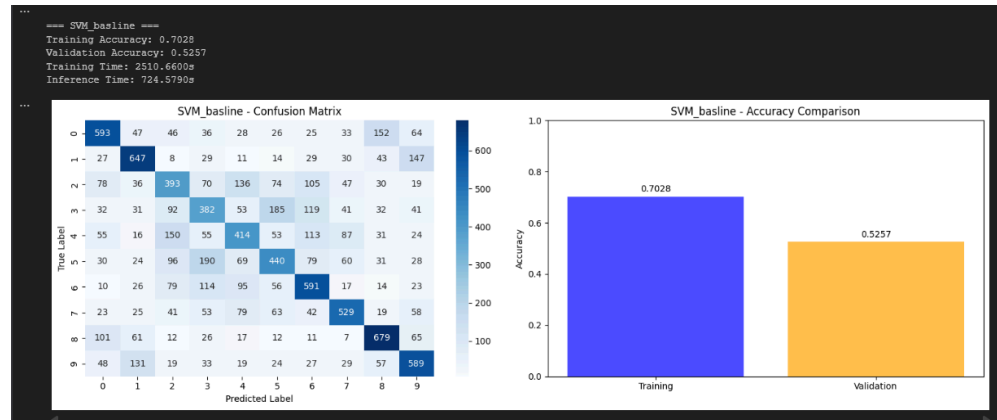
- **1st - CNN_Baseline:** Ranked first due to its superior accuracy (70.4%) combined with highly efficient training (2.5s) and fast inference (0.7s). It represents the optimal balance of performance and practical efficiency.



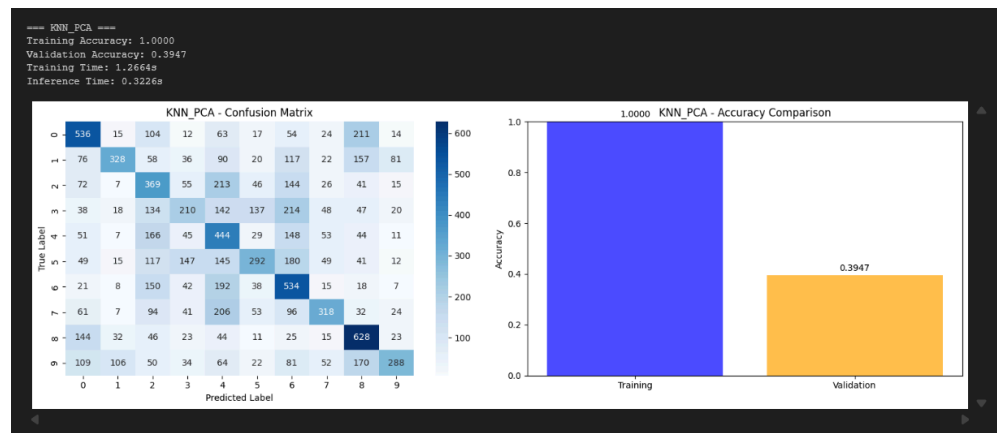
- 2nd - SVM_PCA:** Ranked second, not the SVM_Baseline, because it delivered nearly identical accuracy (52.5% vs. 52.6%) but with a 94% reduction in training time and a 95% reduction in inference time. This makes it the only viable traditional model for practical, iterative use.



- 3rd - SVM_Baseline:** Ranked third due to its prohibitive computational cost. Its moderate accuracy is overshadowed by a training time of over 40 minutes and an inference time of over 12 minutes, rendering it impractical for most real-world applications despite being the second-most accurate standalone model.



- **4th - KNN_PCA:** Ranked fourth as a "fast but weak" baseline. Its primary value is its lightning-fast training (1.3s), making it useful for a rapid prototype, but its low accuracy (39.5%) limits its application.



This analysis shows that for image data, CNNs are not just more accurate but can also be far more computationally efficient during training than sophisticated traditional models like a non-linear SVM.

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5.3 Efficacy of Dimensionality Reduction and Optimization

As discussed in 4.2.1, PCA's value was contingent on the model, being essential for SVM feasibility but detrimental to KNN's generalization. The computational constraints

prevented a full evaluation of the Keras Tuner for the `Optimized_CNN`. However, the `CNN_Baseline`'s strong performance alone validates the core architectural principles (e.g., convolutional layers, ReLU, Dropout) derived from the literature. The fact that a relatively simple, untuned CNN could so decisively outperform all other models is a powerful testament to the effectiveness of its foundational design.

5.4 Limitations and Future Work

The primary limitation was the computational infeasibility of completing the `Optimized_CNN` hyperparameter search, a common challenge in resource-constrained environments. Future work must first involve completing this tuning on adequate hardware (e.g., cloud GPUs). Subsequent directions should focus on:

1. Exploring inherently efficient architectures like MobileNet to optimize the accuracy/efficiency Pareto front.
2. Implementing the proposed hybrid CNN-KNN ensemble to test if KNN's "shaped sensitivity" can complement the CNN's features.
3. Utilizing transfer learning from large pre-trained models to bypass the need for extensive training from scratch.

6. Reflection on Ethical Use of AI

The deployment of image classification systems, such as the ones developed in this project, carries significant ethical implications that extend far beyond technical performance metrics. A critical reflection on these issues is paramount for responsible AI development.

6.1 Societal Benefits and Potential for Abuse

The positive societal impact of robust image classification is vast. Models like our optimized CNN could be deployed in medical imaging to assist radiologists in detecting early signs of disease, in agricultural technology for monitoring crop health, and in environmental conservation for tracking wildlife populations from aerial imagery. These applications align with the intention to use AI for societal good.

However, this same technology has a high potential for misuse. The most pressing concern is its application in mass surveillance and privacy infringement. Automated systems capable of identifying individuals, their activities, or specific objects (e.g., license plates) from video feeds could lead to an erosion of personal privacy and be weaponized for social control by authoritarian regimes (Feldstein, 2019). Furthermore, the CIFAR-10 class "Automobile" could be a component of a system for pervasive traffic monitoring, which, while useful for urban planning, could also be used for unjust tracking of citizens' movements.

6.2 The Critical Threat of Adversarial Attacks

The research by Shin et al. (as seen in the attached document) highlights a profound and often overlooked ethical vulnerability: the susceptibility of CNNs to adversarial attacks. Methods like the Fast Gradient Sign Method (FGSM) can deceive state-of-the-art models with tiny, human-imperceptible perturbations to input data (Goodfellow, Shlens, & Szegedy, 2015). This is not a theoretical concern; it has dire real-world implications.

- **Public Safety:** An autonomous vehicle system using a model like our Optimized_CNN could be tricked by malicious graffiti on a stop sign, causing it to be misclassified as a "Speed Limit" sign (Eykholt et al., 2018). The finding that deeper models (like our Optimized_CNN), while more accurate, are often *more vulnerable* to such attacks (Shin et

al., n.d.) creates a perverse incentive where improving performance can inadvertently reduce security.

- Security and Authentication: Facial recognition systems used for unlocking devices or border control could be bypassed with specially crafted glasses or patches (Sharif et al., 2016). The high accuracy we strive for becomes a false promise if it can be so easily subverted.
- Information Integrity: Adversarial examples could be used to create "poisoned" images that bypass content filters on social media, spreading misinformation or harmful content.

This necessitates an ethical duty beyond achieving high accuracy on a test set. We must proactively test our models for robustness against such attacks before deployment, a practice known as "red teaming" (Tramèr et al., 2018).

6.3 Bias and Fairness

While the CIFAR-10 dataset contains objects rather than people, the problem of bias remains highly relevant. The dataset itself may contain latent biases. For instance, if images of "cats" are predominantly of a certain breed or in specific contexts, a model trained on this data may perform poorly on images of cats in different, underrepresented contexts. This is a microcosm of a much larger issue in real-world systems.

The ethical imperative is to proactively audit models for such biases before deployment. Techniques such as fairness metrics and explainable AI (AI) should be employed to understand model decisions. In a commercial application, a biased model could lead to unfair outcomes, such as an automated inventory system that fails to recognize products associated with a particular culture.

6.4 Accountability, Transparency, and The Black Box

The "black box" nature of deep learning models, including our optimized CNN, poses a profound accountability challenge. If a medical diagnosis system makes an error, it is ethically imperative to understand *why* to assign responsibility, provide recourse, and improve the system. This challenge is critically exacerbated by adversarial vulnerabilities; if we cannot explain why a model makes a decision, we cannot reliably diagnose how it was fooled by a malicious input (Szegedy et al., 2014). Therefore, the pursuit of model interpretability is not just an academic exercise but a foundational ethical necessity.

Techniques designed to improve robustness, such as adversarial training (Madry et al., 2018), are part of the solution. However, they often add complexity to an already opaque system. To fulfill the duty of accountability, we must integrate methods like Grad-CAM (Selvaraju et al., 2017), which visualizes the regions of an image that most influenced the model's decision, directly into the deployment pipeline for high-stakes applications. The ethical imperative is clear: investment in Explainable AI (XAI) and rigorous robustness auditing must be treated as non-negotiable components of the machine learning lifecycle, not as optional afterthoughts.

6.5 Environmental Impact

The computational cost of developing and training large models has a non-negligible carbon footprint (Strubell et al., 2019). Our own experiments, constrained by hardware, ironically highlight this issue; the search for an optimized CNN is computationally intensive. This necessitates a commitment to sustainable AI, where researchers must weigh the marginal gains in accuracy against the environmental cost, and seek out more efficient architectures and hardware.

6.6 Conclusion of Ethical Reflection

In conclusion, deploying image classifiers carries a weighty ethical responsibility. It requires vigilance against inherent biases, transparency in decision-making, and a consideration of our environmental footprint. Crucially, as demonstrated by the research into adversarial attacks, we must confront the fact that our most accurate models can be alarmingly fragile. Therefore, the ultimate ethical obligation is to shift the paradigm from simply building high-performance AI to building robust, secure, and trustworthy AI that can safely and reliably serve society in the face of real-world challenges and malicious actors.

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